Latent Bridge Matching for IDT-Calibrated Style-ID Transfer

Anonymous submission

Abstract

Art-to-art transfer is easy to fake: when the source is already artwork, the unchanged image can score highly for many requested target styles. Raw style-affinity therefore overstates target-style control. We make this failure explicit with identical-image transfer (IDT) calibration on Distinct5-WikiArt and evaluate style movement only above the unchanged-image floor. We then introduce *Latent Bridge Matching* (LBM), a compact style-ID latent transport family trained by endpoint pairing, a style-conditioned latent field, and terminal distribution matching in VAE space. Inference receives only the source image and target style identifier. On Distinct5-WikiArt, IDT calibration reveals a broad failure zone that engulfs reproduced compact baselines, while LBM forms a usable frontier above it. LBM-K clears the calibrated floor after 1.2 minutes on an RTX 3060, the promoted LBM-Knee point reaches 0.7102 transfer CLIP-S at 0.5397 content preservation, and Pattern+Stokes successors push style into the 0.72–0.73 band. Artifact-sensitive diagnostics and a held-out non-CLIP probe separate the closed Knee point from the style-ceiling row. IDT calibration is therefore necessary, and compact latent transport is sufficient to produce a practical style-ID frontier on consumer GPUs.

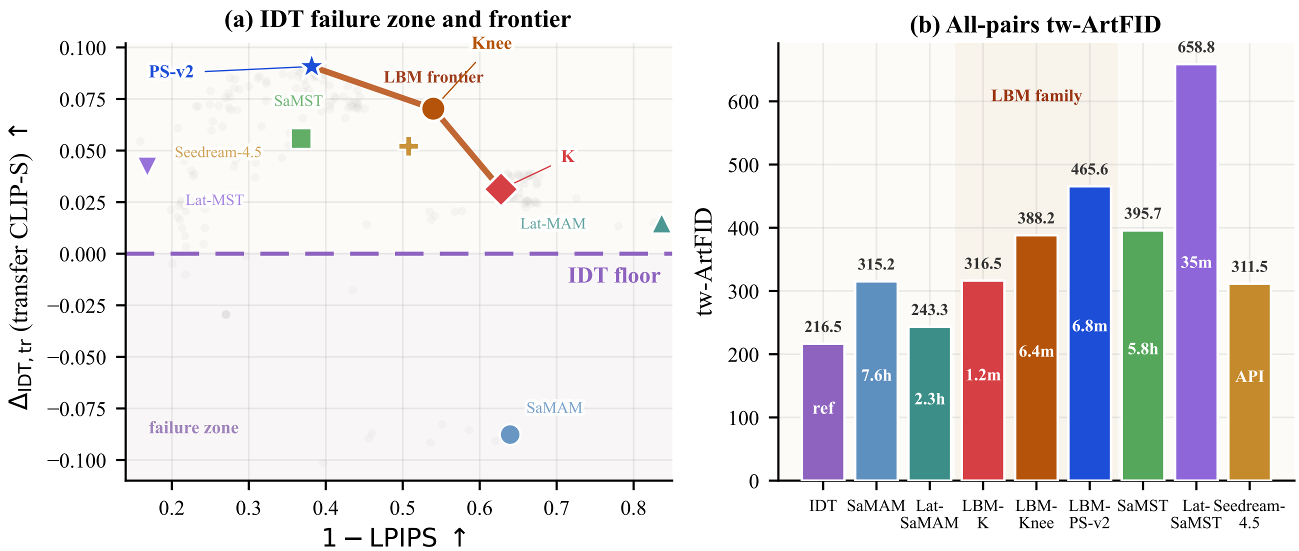

Figure 1: Distinct5-WikiArt is not a single-score benchmark. Panel (a) shows the actual decision surface: a large IDT failure zone below zero style gain, and a usable LBM frontier above it. Gray dots are retained development checkpoints; colored markers are the headline rows, including the latent negatives. Panel (b) keeps the same shortlist under aligned all-pairs target-pooled tw-ArtFID, exposing the artifact cost of that frontier. Train time is printed inside each bar.

Introduction

Domain-level artistic transfer asks for a target *style identity*, not a target exemplar. That setting creates an immediate evaluation failure: when the source image is already artwork, the unchanged image can already score highly for the requested target style. A method can therefore look strong simply by staying inside a broad art manifold. We make this failure explicit with identical-image transfer (IDT) calibration and report only style gain above the unchanged-image floor. Figure 1 shows the result immediately: a large failure zone below IDT and a compact frontier above it.

This regime is stricter than reference-guided stylization and lighter than large-prior editing. Inference receives only a source image and a target style identifier, with no exemplar, per-image optimization, or large-prior adaptation. The question is whether a compact style-ID model can still produce reliable target-style motion under that contract.

We answer yes with *Latent Bridge Matching* (LBM), a compact VAE-space transport family built from training-side endpoint pairing, a style-conditioned latent field, and terminal distribution matching with kinetic control. On Distinct5-WikiArt, LBM forms a real frontier: minute-scale anchors clear the IDT floor, LBM-Knee closes the strongest style-cleanliness tradeoff, and Pattern+Stokes successors extend the style ceiling. The paper makes three concrete claims: IDT calibration is necessary; compact style-ID transport works without exemplars or large-prior adaptation; and Distinct5-WikiArt exposes a measurable frontier rather than a single winner.

Related Work

Most style-transfer systems solve a different inference contract: they either consume a target exemplar or inherit a strong pretrained generator at test time (Gatys, Ecker, and Bethge 2016; Huang and Belongie 2017; Deng et al. 2022; Hong et al. 2023; Huang et al. 2024; Zhang et al. 2024; Chung, Hyun, and Heo 2024). Our setting keeps only the target style identity, placing it closest to compact multi-style baselines such as SaMST, Mamba-ST, and SaMam (Liu et al. 2024; Botti et al. 2025; Liu et al. 2025). The nearest transport-side analogy is low-energy one-step editing (Lu et al. 2026), although that line targets prompt-conditioned semantic editing rather than domain-level style-ID transfer.

Evaluation is also different. Automatic stylization metrics are unstable and only partially aligned with human judgment (Zhou et al. 2024; Yeh et al. 2020; Wright and Ommer 2022), and in art-to-art transfer they reward the unchanged image far too easily. We therefore make the no-op floor explicit and report Δ_{idt} before broader plausibility or artifact diagnostics. Full related-work detail is deferred to the supplement.

Method

Inference contract

LBM is an endpoint transport model in VAE latent space for domain-level artistic transfer. Figure 1 established the target: clear the IDT floor without paying the artifact cost of the strongest high-style baselines. Figure 2 shows the contract used to trace that frontier: style-ID control on top, the active latent transport path in the middle, and endpoint construction plus terminal matching only on the training side. For Distinct5-WikiArt, the source latent is $z_0 \in \mathbb{R}^{4\times64\times64}$; the historical CycleGAN-256 packet uses $4 \times 32 \times 32$ latents. At inference, the model receives only the source image and target style identifier. Target-domain latents remain training-side supervision only.

Our OT view is deliberately limited and practical. As in recent low-energy transport views of one-step editing (Lu et al. 2026), OT is used as supervisory geometry, not as an online solver claim. In the reproduced Distinct5 mainline, the active headline rows use endpoint-mode OMF rather than the earlier random-time flow residual:

$$\hat{v} = v_\theta(z_0, 1, s), \qquad \hat{z}_1 = z_0 + \hat{v}, \tag{1}$$

with active objective

$$\mathcal{L}_{\text{active}} = \lambda_{\text{term}}\mathcal{L}_{\text{term}}(\hat{z}_1) + \lambda_{\text{kin}}\|\hat{v}\|_2^2, \tag{2}$$

where the pairing cache selects terminal targets and the parent flow residual is disabled in the headline rows ($w_{\text{flow}} = 0$). We therefore describe the headline Distinct5 family as pairing-cache / terminal-SWD / kinetic OMF.

Training-side endpoint construction and latent execution

We train directly on precomputed VAE latents without paired supervision. For each source latent z_0 and target style s, the pairing cache exposes a selector-restricted candidate set $\mathcal{C}(z_0, s)$ from the target domain. The terminal target $\tilde{z}_1$ is drawn from that scheduled set, and the successor family stabilizes it with barycentric targets and queue-side target smoothing. Endpoint pairing therefore decides *where* the model should move; the latent field still learns *how* to execute that move.

Execution is handled by a time-conditioned latent backbone with sinusoidal time embeddings, style spatial priors, and semantic cross-attention. The tokenizer receives only the target style identifier and produces a compact control code c_s; the style-conditioned latent field then decides how much motion is needed under the current content.

The endpoint is then regularized by semantic-aligned sliced Wasserstein distance. Let $\Phi_\theta(z_0, s)$ denote the Euler-integrated endpoint. The terminal objective is

$$\mathcal{L}_{\text{term}} = \text{SWD}_1\big(\Phi_\theta(z_0, s),\ \tilde{z}_1\big), \tag{3}$$

where sorted one-dimensional projections compare generated endpoint patches against target-style patches without requiring spatial correspondence. In the current SA-SWD instantiation, projection directions are chosen from the same routing signal that injects style into the backbone, keeping target-style pressure distributional rather than pointwise. The supplement gives bounded design-grounding support for the active field/solver regime: a local path-action justification for endpoint kinetic regularization and the expected $O(1/K)$ Euler discretization bound.

Operating-point variants. All reported rows share the same test-time contract. **LBM-K** is the kinetic-only top-2 anchor; **LBM-Knee** adds top-8 barycentric + dual-target supervision, anisotropic + Stokes control, and auxiliary terminal SWD; **LBM-PS** and **LBM-PS-v2** keep the same successor family and differ only in the Stokes coefficient $(0.05/0.02)$, forming the balanced high-style and style-ceiling rows used later in the frontier analysis.

Experiments

Experimental setup

We report one main benchmark and one historical secondary packet. Distinct5-WikiArt carries the main claim; the older CycleGAN-256 packet is kept only as secondary support and is never merged into the same leaderboard. Figure 1 asks whether a compact style-ID family can move above the IDT floor while staying in a usable artifact regime; the rest of this section measures where each row lands on that surface.

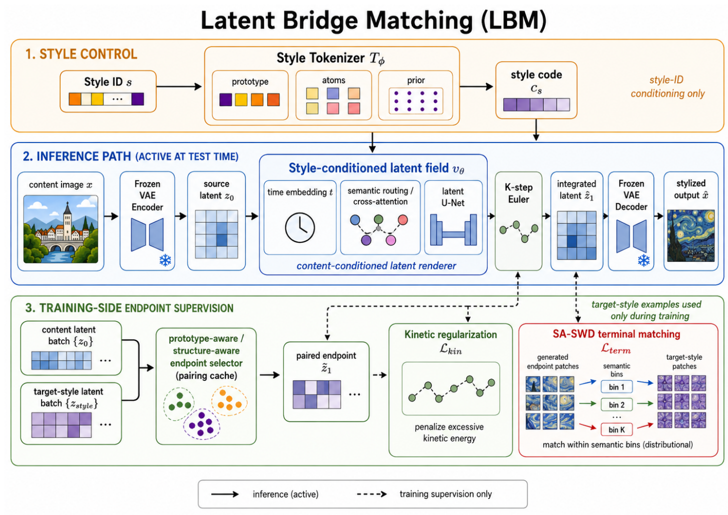

Figure 2: LBM in three bands. The top band contains style-ID control only. The middle band is the active test-time latent transport path that must move above the Figure 1 failure zone and into the frontier. The bottom band contains endpoint construction, kinetic control, and SA-SWD, all used only during training.

Dataset and evaluation metrics. Distinct5-WikiArt uses Early Renaissance, Impressionism, Minimalism, Rococo, and Ukiyo-e, with 1000 training images and 30 held-out test images per class. The classes are chosen from WikiArt by a CLIP-style separation screen, and evaluation again uses all 5×5 directions over 750 generated images.

The main metrics are transfer CLIP-S, LPIPS, and

$$\mathrm{EC} = \text{CLIP-style} \cdot (1 - \text{LPIPS}). \tag{4}$$

For Distinct5-WikiArt we additionally report style gain over the unchanged-image reference,

$$\Delta_{\mathrm{idt}} = \text{CLIP-S(method)} - \text{CLIP-S(idt)}, \tag{5}$$

under both the full 5×5 protocol and the transfer-only off-diagonal subset. On Distinct5, Δ_{idt} is the first sanity check: any row at or below zero has failed to move reliably toward the requested target style. EC is used only as a Pareto summary. Auxiliary diagnostics are targetwise ArtFID, a held-out non-CLIP style probe, and standard perceptual/frequency checks (Ke et al. 2021; Yang et al. 2022; Ding et al. 2020; Bińkowski et al. 2018). The historical CycleGAN-256 packet follows the same 750-output protocol and is used only as secondary support later in the paper.

Implementation details. All paper-facing LBM rows are trained and evaluated on the reviewed single-lane RTX 3060 WSL surface, and reported train time is cumulative wall-clock to the shown checkpoint with evaluation excluded. Same-cost comparisons therefore align training wall-clock rather than mixed train+eval runtime. Seedream-4.5 is kept only as an external large-prior reference under a fixed API prompt protocol, not as a same-interface style-ID baseline.

Main result on Distinct5-WikiArt

Figure 1 and Table 1 give the paper's main result. Distinct5-WikiArt is an IDT-calibrated failure test, not a single-score leaderboard. The unchanged-image control already reaches CLIP-S 0.6801, so any row with $\Delta_{\mathrm{idt}} \le 0$ fails before trade-off analysis begins.

Quantitative frontier. LBM occupies three regimes on that surface: minute-scale anchors, a content-preserving Knee point, and a stronger style band. LBM-K reaches 0.6712 / 0.6277 in transfer CLIP-S / $(1-\text{LPIPS})$, the promoted Knee point reaches 0.7102 / 0.5397, and the stronger successors extend the style side to 0.7274 / 0.3967 and 0.7307 / 0.3817. This is a frontier, not a single winner.

Baseline bounds. The reproduced baselines bound that frontier from both sides. SaMST e15 reaches 0.6957 / 0.6319, but in a much higher-damage region, and its retained e5/e15 packet is already near a plateau. Seedream-4.5 is a stronger external large-prior reference, but its transfer row, 0.6920 / 0.4923, still stays below LBM-Knee on both style gain and content preservation. SaMAM-2250 never clears the IDT floor. The latent negatives sharpen the same failure mode: Lat SaMAM stays only +0.0148 above IDT, whereas Lat SaMST collapses to the worst all-pairs ArtFID band at 658.8.

Table 1: Selected Distinct5-WikiArt transfer points. 'idt' is the unchanged-image floor. The ordering is deliberate: floor, failed compact baseline, reproduced high-style baselines, compact LBM anchor, closed Knee point, and stronger promoted successors.

Point	Role	CLIP-S$_{\mathrm{tr}}$ ↑	$1 -$ LPIPS ↑	$\Delta_{\mathrm{idt,tr}}$ ↑	Closure
idt	no-op floor	0.6399	1.0000	0.0000	closed
SaMAM-2250	compact baseline	0.5523	0.6395	-0.0877	closed
SaMST e15	high-style baseline	0.6957	0.3681	+0.0558	closed
Seedream-4.5	large-prior ref.	0.6920	0.5077	+0.0521	closed
LBM-K e1	conservative anchor	0.6712	0.6277	+0.0312	closed
LBM-Knee e13	content-preserving successor	0.7102	**0.5397**	+0.0703	closed
LBM-PS e13	balanced high-style	0.7274	0.3967	+0.0875	frontier
LBM-PS-v2 e13	style ceiling	**0.7307**	0.3817	**+0.0908**	frontier

Table 2: Artifact-sensitive and non-CLIP diagnostics on selected Distinct5-WikiArt rows. tw-ArtFID uses the all-pairs target-pooled recompute and separates the closed Knee point from the style-ceiling and large-prior rows.

Point	tw-ArtFID$_{\mathrm{all}}$ ↓	Gram-μ ↓	EdgePurity↑	NonCLIPAcc↑
LBM-K e1	316.5	0.153	**0.540**	0.167
LBM-Knee e13	388.2	**0.137**	0.288	0.237
LBM-PS-v2 e13	465.6	0.183	0.115	0.272
SaMST e15	395.7	0.147	0.102	0.248
Seedream-4.5	**311.5**	0.189	0.389	**0.378**

Artifact-sensitive diagnostics. Table 2 explains why Knee is the promoted default. Against SaMST e15, Knee lowers all-pairs target-pooled ArtFID (388.2 vs. 395.7), improves Gram-μ (0.137 vs. 0.147), and more than doubles edge purity (0.288 vs. 0.102). The held-out non-CLIP probe keeps Knee close to SaMST (0.237 vs. 0.248), while PS-v2 and Seedream move higher on style. Bootstrap keeps both promoted rows safely above IDT, so Knee remains the main closed point and PS-v2 the explicit style ceiling.

Qualitative read. Figure 3 shows the same pattern visually. Panel A makes the calibration point concrete: plausible outputs can still miss the requested style move. Panel B shows the grouped frontier itself: high-style baselines and PS-v2 push harder toward the target domain, while LBM-Knee keeps more of the source layout and still moves beyond the compact anchor.

Table 3: Distinct5-WikiArt recorded operating-point cost under the reviewed full 5×5 / 750-output packet. Reported train time is cumulative training wall-clock to the shown checkpoint, with evaluation excluded. This is recorded operating-point cost, not a normalized time-to-quality comparison.

Method	Recorded point	Train to point
LBM-F	e1	**1.2 min**
LBM-K	e1	**1.2 min**
SaMAM	step 2250	7.6 h
SaMST	e15	5.8 h

Note. Distinct5 times are cumulative training wall only, with evaluation excluded. We intentionally omit a shared inference column because same-grade pure-generation timing is still incomplete for the active manuscript rows `SaMAM-2250` and `SaMST-e15`; leaving blanks would overstate closure. Post-2250 SaMAM timings are intentionally excluded.

Retest and cost. The same ordering survives direct reevaluation and cost comparison. Under explicit three-scope retest, Knee reaches 0.7101 / 0.4604 on transfer and 0.7302 / 0.4560 on all-pairs in CLIP-S / LPIPS, versus 0.6920 / 0.4923 and 0.7198 / 0.4767 for Seedream-4.5. Table 3 then shows why that row matters in practice: conservative LBM anchors cross the Distinct5-WikiArt IDT floor after 1.2 minutes on a single RTX 3060, whereas SaMAM-2250 remains below the floor after 7.6 hours and SaMST exceeds it only after 5.8 hours in a much higher-damage region. The family is therefore selectable rather than singular: LBM-K is the conservative anchor, LBM-Knee the default closed point, and PS-v2 the style-ceiling row.

Historical support and secondary evidence

We keep the historical CycleGAN-256 packet only as secondary support. On that older surface, LBM remains a competitive compact retrainable point at 0.7161 CLIP-S / 0.4514 LPIPS and 114 ms/image, while the strongest reproduced competitors either collapse content or enter noisier artifact regimes. Two fixed-rule WikiArt follow-up splits preserve the same evaluator pattern: unchanged images still score strongly, and compact LBM-F remains above the split-local IDT floor by ‘+0.0139‘ and ‘+0.0073‘ transfer CLIP-S. The supplement carries the artifact ledger and protocol details.

Discussion and Limitations

The main lesson is not that one row wins every metric; it is that Distinct5-WikiArt is easy to misread. Raw CLIP-style overstates success in art-to-art transfer. Once the unchanged-image floor is explicit, several reproduced baselines stop counting as evidence of target-style control: some never beat the unchanged image, while others do so only by entering substantially worse LPIPS and artifact regions. The benchmark then exposes a compact frontier: low-damage anchors, a closed Knee point, and a stronger style-ceiling regime.

The limitations are direct. The task is domain-level rather than exemplar-level, Euler integration is intentionally simple, and artifact-sensitive diagnostics are still proxies rather than preference judgments. Distinct5-WikiArt is a stress benchmark rather than a universal art benchmark, although the fixed-rule follow-up splits reproduce the same unchanged-image pathology.

Conclusion

IDT calibration changes what counts as success in domain-level art transfer. Once the unchanged-image prior is measured explicitly, compact latent transport is enough to produce useful style-ID operating points: minute-scale anchors, a closed Knee point, and a stronger style-ceiling regime. LBM therefore offers a practical small-model alternative to large-prior adaptation on Distinct5-WikiArt.

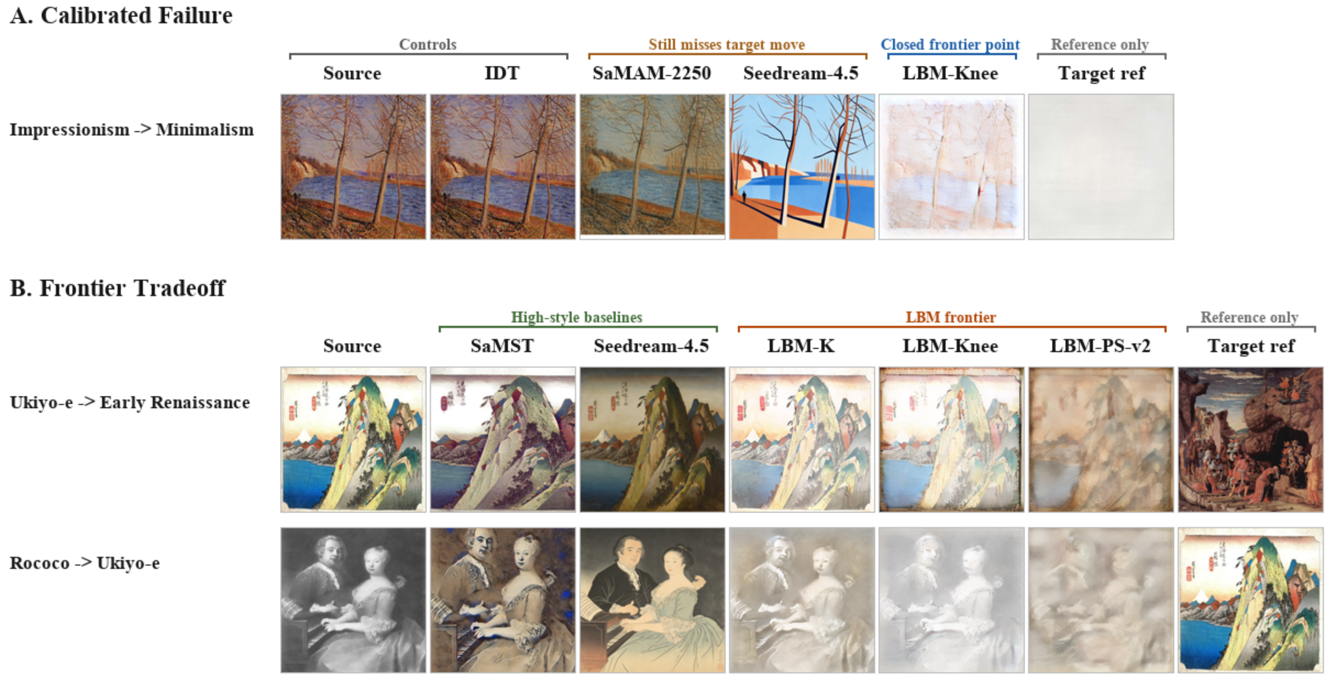

Figure 3: Qualitative reading of Distinct5-WikiArt. Columns are grouped into controls, reproduced baselines, and frontier rows. Panel A shows the calibrated failure mode on 'Impressionism $\rightarrow$ Minimalism': plausible outputs can still miss the requested target-style move. Panel B shows the same frontier tradeoff as Figure 1: stronger style movement pulls toward SaMST / Seedream / PS-v2, while Knee keeps more of the source layout. Target-domain references are shown only for visualization.

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Reproducibility Checklist

This paper

- Includes a conceptual outline and/or pseudocode description of AI methods introduced: **Yes**.
- Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results: **Yes**.
- Provides well marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper: **Yes**.

Does this paper make theoretical contributions? Yes (design-grounding). The paper provides compact formal analysis with proofs and empirical checks: Theorem 1 and Theorem 2 analyze the active field/solver regime through a path-action decomposition for kinetic regularization and an $O(1/K)$ Euler discretization bound, while Theorem 3 analyzes a historical OT-style assignment variant as family-level background. Full proofs are in the supplement. These results justify the three-part design without asserting equivalence to a full Schrödinger Bridge solver.

Does this paper rely on one or more datasets? Yes.

- A motivation is given for why the experiments are conducted on the selected datasets: **Yes**.
- All novel datasets introduced in this paper are included in a data appendix: **NA**.
- All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes: **NA**.
- All datasets drawn from the existing literature are accompanied by appropriate citations: **Yes**.
- All datasets drawn from the existing literature are publicly available: **Yes**.
- All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisficing: **NA**.

Does this paper include computational experiments? Yes.

- This paper states the number and range of values tried per (hyper-)parameter during development of the paper, along with the criterion used for selecting the final parameter setting: **Partial**.
- Any code required for pre-processing data is included in the appendix: **Partial** (the supplement lists the reviewed script paths and artifact roots for the current paper-facing packets).
- All source code required for conducting and analyzing the experiments is included in a code appendix: **Partial** (the supplement enumerates the main reviewed scripts, configs, and eval artifacts, but does not inline the full codebase).
- All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes: **Partial**.
- All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from: **Partial**.
- If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results: **Partial** (the supplement records the reviewed seed settings for the current paper-facing packets, but external references such as Seedream do not expose a user-controlled seed).
- This paper specifies the computing infrastructure used for running experiments, including hardware/software details: **Yes**.
- This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics: **Yes**.
- This paper states the number of algorithm runs used to compute each reported result: **Yes**.
- Analysis of experiments goes beyond single-dimensional summaries of performance to include measures of variation, confidence, or other distributional information: **Yes**.
- The significance of any improvement or decrease in performance is judged using appropriate statistical tests: **Partial** (selected confidence intervals and bootstrap summaries are reported, but not every manuscript comparison is presented as a closed significance claim).
- This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments: **Partial**.